Projections onto Subspaces

When you can't hit a target or hedge a risk exactly, get as close as possible. The closest point is found by dropping a perpendicular. That projection is the best possible hedge, the hedge ratio it gives is beta, and what's left over is the risk no hedge can touch.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. **Conditions and important concepts**
   12 short definitions: projection onto a line and a subspace, the error, the projection matrix, the normal equations.
2. **Procedure**
   Project onto a line; project onto a subspace.
3. **2 × 2 example**
   Hedge a fund with Nifty in two scenarios; P and $P^2 = P$.
4. **3 × 3 example: the hedge ratio is beta**
   Midcap = 1.25 × Nifty + a market-neutral leftover. With a hedge-ratio slider.
5. **Bigger example: the closest reachable target**
   The target (10, 105, 4) is impossible; the normal equations find the nearest one.
6. **Connection: why projections matter**
   Best hedge, least squares, and the four subspaces.
7. **Final memory table**
   Every formula on one screen.

METHODS IN THIS LESSON: Projection onto a line | Projection matrix aa^T/a^Ta | $P^T = P$, $P^2 = P$ | Normal equations $A^TA\hat{x} = A^Tb$ | $P = A(A^TA)^{-1}A^T$ | Best hedge = projection

PART 2 Prerequisites

Click any link to review it, then come back.

- Orthogonality, length, A^TA (Lesson 13).
- The plane of reachable targets and the hidden link (Lesson 10).

- The 2×2 inverse (Lesson 3).

PART 3

Explanation

THE NUMBERS USED IN THE EXAMPLES

Demeaned scenario returns (bad, normal, good year) from Lesson 13: Nifty $a = (-20, 0, 20)$, midcap $b = (-26, 2, 24)$. Fund matrix with Nifty and bonds and three rules (budget, return, exposure): columns $(1, 12, 1)$ and $(1, 7, 0)$.

1. Conditions and important concepts

1. The question

Given b and a line (or plane) of things you can reach, which reachable point is closest to

Picture it: drop a perpendicular from b onto the line. Where it lands is the closest point p, the **projection**. The gap $e = b - p$ is the **error**.

2. Projection onto a line

METHOD · PROJECTION ONTO THE LINE THROUGH A

p is on the line, so $p = \hat{x}\, a$ for some number $\hat{x}$. The error must be perpendicular to a:

$$a^T(b - \hat{x}\, a) = 0 \;\Rightarrow\; \hat{x}\, a^T a = a^T b \;\Rightarrow\; \boxed{\hat{x} = \frac{a^T b}{a^T a}} \qquad p = \hat{x}\, a$$

3. The error is perpendicular

$e = b - p$ satisfies $a^T e = 0$. That is **why** p is closest: any other point $\hat{x}'a$ on the line gives $b - \hat{x}'a = e + (\hat{x} - \hat{x}')a$, and by Pythagoras its length is $\|e\|^2 + (\text{something})^2 \geq \|e\|^2$.

4. What scaling does

Doubling b doubles p. Doubling a leaves p unchanged (it's the same line): $\hat{x}$ halves while a doubles.

5. The projection matrix (line)

METHOD · PROJECTION MATRIX ONTO A LINE

$$p = a\,\frac{a^T b}{a^T a} = \underbrace{\frac{a a^T}{a^T a}}_{P}\, b \qquad P = \frac{a a^T}{a^T a}$$

aa^T is a matrix (a column times a row), not a number. P has rank 1: its column space is the line through a.

6. Two properties of every projection

RULE · $P^T = P$ AND $P^2 = P$

P is symmetric. And projecting twice changes nothing: once b is on the line, its projection is itself, so $P^2 = P$.

7. Why project?

$Ax = b$ often has no solution: b isn't in $C(A)$ (Lesson 8). So replace b by the closest vector that **is** in $C(A)$, its projection p, and solve $A\hat{x} = p$ instead.

8. Projection onto a subspace: the normal equations

METHOD · THE NORMAL EQUATIONS

Let the columns of A span the subspace. Then $p = A\hat{x}$, and the error $e = b - A\hat{x}$ must be perpendicular to every column of A:

$$A^T(b - A\hat{x}) = 0 \;\Rightarrow\; \boxed{A^T A\,\hat{x} = A^T b}$$

With one column, $A^T A = a^T a$ is a number, and this is concept 2 again.

9. The formulas

METHOD · PROJECTION ONTO C(A)

$$\hat{x} = (A^T A)^{-1} A^T b \qquad p = A\hat{x} \qquad P = A(A^T A)^{-1} A^T$$

$A^T A$ is invertible because the columns are independent (Lesson 13). Don't "simplify" $(A^T A)^{-1}$ into $A^{-1}(A^T)^{-1}$: A is usually not square.

10. Where the error lives

$A^T e = 0$ says e is in the **left null space** $N(A^T)$, which is perpendicular to $C(A)$ (Lesson 13). So b splits into two perpendicular pieces:

$$b = \underbrace{p}_{\in C(A)} + \underbrace{e}_{\in N(A^T)}$$

11. I – P

$I - P$ is also a projection (MIT problem 15.2: $(I-P)^2 = I - 2P + P^2 = I - P$). It projects onto the left null space and gives the error: $(I-P)b = e$.

12. In markets: the best hedge

> **MODEL · THE BEST HEDGE IS A PROJECTION**
>
> To hedge returns b with an instrument whose returns are a, choose the hedge ratio h that makes the leftover risk $\|b - ha\|^2$ as small as possible. The answer is the projection: $h = \dfrac{a^T b}{a^T a}$, which for demeaned returns is **beta** (covariance ÷ variance). The leftover e is uncorrelated with the hedge: the **unhedgeable** part.

2. Procedure

A. Onto a line through a

Step	What to do
1	Compute $a^T b$ and $a^T a$.
2	$\hat{x} = a^T b / a^T a$; $p = \hat{x}\, a$.
3	$e = b - p$. Check $a^T e = 0$.
4	(Optional) $P = aa^T / a^T a$; check $Pb = p$.

B. Onto the column space of A

Step	What to do
1	Compute $A^T A$ and $A^T b$.
2	Solve $A^T A\,\hat{x} = A^T b$.
3	$p = A\hat{x}$; $e = b - p$.
4	Check $A^T e = 0$ (the error is perpendicular to every column).

3. 2 × 2 example

Two scenarios (bad year, good year). Nifty returns $a = (-10, 20)$; a large-cap fund returns $b = (-11, 25)$. Best hedge ratio?

1 DOT PRODUCTS $a^T b = (-10)(-11) + (20)(25) = 110 + 500 = 610.$ $a^T a = 100 + 400 = 500.$

RATIO $\hat{x} = 610/500 = 1.22.$

2 PROJECTION $p = 1.22 \times (-10, 20) = (-12.2, 24.4).$

ERROR $e = (-11 + 12.2,\ 25 - 24.4) = (1.2, 0.6).$

CHECK $a^T e = (-10)(1.2) + (20)(0.6) = -12 + 12 = 0$ ✓.

Projection matrix:

$$P = \frac{aa^T}{a^T a} = \frac{1}{500}\begin{bmatrix} 100 & -200 \\ -200 & 400 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.4 \\ -0.4 & 0.8 \end{bmatrix}$$

Check Pb: $(0.2(-11) - 0.4(25),\ -0.4(-11) + 0.8(25)) = (-2.2 - 10,\ 4.4 + 20) = (-12.2, 24.4) = p$ ✓. Check P^2: top-left $0.04 + 0.16 = 0.2$, top-right $-0.08 - 0.32 = -0.4$, bottom-right $0.16 + 0.64 = 0.8$: $P^2 = P$ ✓.

Meaning: short ₹1.22 of Nifty for every ₹1 of the fund. The leftover $(1.2, 0.6)$ is risk Nifty can't cancel.

4. 3 × 3 example: the hedge ratio is beta

Three scenarios, demeaned returns: Nifty $a = (-20, 0, 20)$, midcap $b = (-26, 2, 24)$.

1 DOT PRODUCTS $a^T b = 520 + 0 + 480 = 1000.\ a^T a = 400 + 0 + 400 = 800.$

BETA $\hat{x} = 1000/800 = \boxed{1.25}.$

2 PROJECTION $p = 1.25 \times (-20, 0, 20) = (-25, 0, 25)$: the part of midcap that the market explains.

ERROR $e = (-26 + 25,\ 2 - 0,\ 24 - 25) = (-1, 2, -1)$: the part it doesn't.

CHECK $a^T e = 20 + 0 - 20 = 0$ ✓: the leftover is uncorrelated with Nifty.

$$\boxed{\text{midcap} = 1.25 \times \text{Nifty} + (-1, 2, -1)}$$

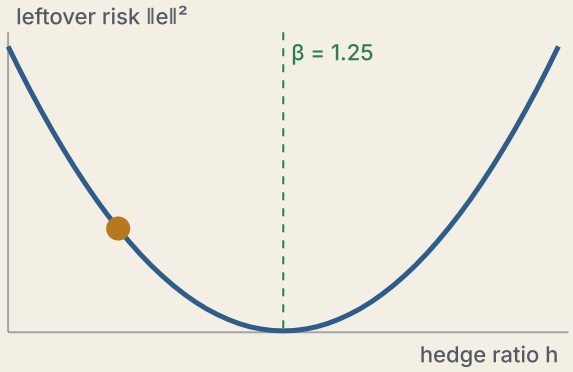
And $(-1, 2, -1) = -0.2 \times (5, -10, 5)$: the leftover is Lesson 13's market-neutral strategy, scaled. **Hedge:** for every ₹1 of midcap, short ₹1.25 of Nifty. What remains can't be hedged with Nifty, and it's small: $\|e\|^2 = 1 + 4 + 1 = 6$ against $\|b\|^2 = 1256$.

The projection matrix onto the Nifty line:

$$P = \frac{aa^T}{800} = \frac{1}{800}\begin{bmatrix} 400 & 0 & -400 \\ 0 & 0 & 0 \\ -400 & 0 & 400 \end{bmatrix} = \begin{bmatrix} 0.5 & 0 & -0.5 \\ 0 & 0 & 0 \\ -0.5 & 0 & 0.5 \end{bmatrix} \qquad Pb = \begin{bmatrix} -13 - 12 \\ 0 \\ 13 + 12 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 25 \end{bmatrix}$$

Try it: find the best hedge ratio yourself

Hedge ratio h (₹ of Nifty shorted per ₹1 of midcap) 0.50



Leftover $e = b - h\,a = (-16, 2, 14)$

Risk $\|e\|^2 = 456$. Check $a^T e = 600$.

Under-hedged: still positively tied to Nifty ($a^T e > 0$). Slide toward 1.25.

5. Bigger example: the closest reachable target

With Nifty and bonds, reachable targets must satisfy $R = 7B + 5E$ (Lesson 10). A client asks for $b = (10, 105, 4)$: ₹10 lakh, ₹105 thousand, exposure 4. Test: $70 - 105 + 20 = -15 \neq 0$. Impossible. What's the **closest** reachable target?

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \\ 1 & 0 \end{bmatrix} \qquad b = \begin{bmatrix} 10 \\ 105 \\ 4 \end{bmatrix}$$

Step 1, $A^T A$ and $A^T b$ (each entry a dot product of columns):

$$A^T A = \begin{bmatrix} 1+144+1 & 1+84+0 \\ 1+84+0 & 1+49+0 \end{bmatrix} = \begin{bmatrix} 146 & 85 \\ 85 & 50 \end{bmatrix} \qquad A^T b = \begin{bmatrix} 10+1260+4 \\ 10+735+0 \end{bmatrix} = \begin{bmatrix} 1274 \\ 745 \end{bmatrix}$$

Step 2, solve $A^T A\hat{x} = A^T b$. Determinant $146 \times 50 - 85 \times 85 = 7300 - 7225 = 75$:

$$\hat{x} = \frac{1}{75}\begin{bmatrix} 50 & -85 \\ -85 & 146 \end{bmatrix}\begin{bmatrix} 1274 \\ 745 \end{bmatrix} = \frac{1}{75}\begin{bmatrix} 63700 - 63325 \\ -108290 + 108770 \end{bmatrix} = \frac{1}{75}\begin{bmatrix} 375 \\ 480 \end{bmatrix} = \begin{bmatrix} 5 \\ 6.4 \end{bmatrix}$$

Step 3, the closest target and the error:

$$p = A\hat{x} = \begin{bmatrix} 5 + 6.4 \\ 60 + 44.8 \\ 5 + 0 \end{bmatrix} = \begin{bmatrix} 11.4 \\ 104.8 \\ 5 \end{bmatrix} \qquad e = b - p = \begin{bmatrix} -1.4 \\ 0.2 \\ -1 \end{bmatrix}$$

Step 4, checks. $A^T e$: column 1 $-1.4 + 2.4 - 1 = 0$ ✓, column 2 $-1.4 + 1.4 + 0 = 0$ ✓. And p is reachable: $7(11.4) + 5(5) = 79.8 + 25 = 104.8$ ✓.

Meaning. The best you can do with Nifty and bonds is ₹5 lakh Nifty + ₹6.4 lakh bonds, giving (₹11.4L, ₹104.8K, exposure 5). The error $(-1.4, 0.2, -1) = -0.2 \times (7, -1, 5)$ points along the hidden link: it lives in the left null space, exactly as concept 10 says.

▼ For the curious: a shortcut, and a caution about units

Because the plane's normal is $n = (7, -1, 5)$, you can also project by removing the normal component: $p = b - \frac{n^T b}{n^T n} n = b - \frac{-15}{75} n = b + 0.2(7, -1, 5)$. Same answer. And $P = I - \frac{nn^T}{n^T n}$.

Caution: "closest" here treats ₹1 lakh of budget, ₹1 thousand of return and 1 unit of exposure as equally important. In practice you'd weight the rules (weighted least squares, coming up) by how much each miss matters.

6. Connection: why projections matter

Setting	b	Subspace	Projection p	Error e
Hedging with one instrument	a fund's returns	line of the hedge	hedged part (beta × hedge)	unhedgeable risk
Hedging with several	an exposure / payoff	$C(A)$ of the hedges	best replicating hedge	residual risk, in $N(A^T)$
Unreachable target	client's target	reachable targets $C(A)$	closest reachable target	along the hidden link
Fitting data (Lesson 15)	observed points	all lines $C + Dt$	best-fit line	residuals

The same picture every time: $b = p + e$ with $p \in C(A)$, $e \in N(A^T)$, $p \perp e$. Next lesson uses it for **least squares**: fitting a line to market data, like estimating beta and alpha from returns.

7. Final memory table

	Onto a line (through a)	Onto a subspace ($C(A)$)
Coefficient	$\hat{x} = a^T b / a^T a$	$A^T A\hat{x} = A^T b$
Projection	$p = \hat{x}\, a$	$p = A\hat{x}$
Matrix	$P = aa^T / a^T a$ (rank 1)	$P = A(A^T A)^{-1} A^T$
Error	$e = b - p,\ a^T e = 0$	$e = b - p,\ A^T e = 0,\ e \in N(A^T)$
Properties	$P^T = P,\ P^2 = P$; $I - P$ projects onto the perpendicular complement	
Markets	hedge ratio = beta	best hedge; closest reachable target

▼ **Check yourself (MIT problem 15.1): A = the 4 × 4 identity with its last column removed. Project $b = (1, 2, 3, 4)$ onto $C(A)$.**

$C(A)$ is all vectors with last entry 0. So $P = \text{diag}(1, 1, 1, 0)$ (4 × 4) and $p = Pb = (1, 2, 3, 0)$. The error $(0, 0, 0, 4)$ is in the left null space, and $I - P = \text{diag}(0, 0, 0, 1)$ projects onto it.

▼ **Check yourself (MIT recitation): the projection matrix onto the plane $x + y - z = 0$.**

Basis $(1, -1, 0)$, $(1, 0, 1)$; $A^TA = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, inverse $\frac{1}{3}\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, so $P = \frac{1}{3}\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$. Check: $P(1, 1, -1) = 0$ (the normal goes to zero). Shortcut: $P = I - \frac{nn^T}{n^Tn}$ with $n = (1, 1, -1)$.